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Question 1 [C01] [10 Points]

In the wizarding world of Harry Potter, Gringotts Bank has a **magical vault** that protects treasures using a secret spell. Harry has been given a mysterious number by the goblins, and he must calculate its **Vault Value** to unlock the treasure. The rules of the vault are enchanted as follows: If a digit is **even**, the vault considers it **too heavy**, and its **square** is **subtracted** from the Vault Value. If a digit is **odd and less than or equal to 3**, the vault considers it **magically potent**, and its **cube** is **added** to the Vault Value. If a digit is **divisible by 3**, the vault recognizes it as a **lucky number** and **adds** the digit itself to the Vault Value. All other digits are **cursed**, and their value is **subtracted** from the Vault Value.

Harry must process the **entire number using only one loop** – a requirement imposed by the ancient goblins' magic.

Write a program that takes a number from Harry(user) and calculates the **Vault Value** according to the above magical rules. This Vault Value will tell him if he can safely open the vault and retrieve the treasure. [You cannot use string, array here]

Input	Output
code_number = 46385	vault_value= -94
<u>Explanation:</u> Last digit = 5, Vault Value change: $0 - 5 = -5$ Next digit = 8, Vault Value change: $-5 - (8^2) = -5 - 64 = -69$ Next digit = 3, Vault Value change: $-69 + (3*3*3) = -69 + 27 = -42$ Next digit = 6, Vault Value change: $-42 - (6^2) = -42 - 36 = -78$ Next digit = 4, Vault Value change: $-78 - (4^2) = -78 - 16 = -94$	